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Fluid flushing system

Abstract

A fluid flushing system includes a sampling port comprising a sampling conduit configured to dispense a sample of a fluid in the fluid routing assembly; a valve configurable in an open valve configuration, wherein the fluid is permitted to flow through the fluid flushing system, and a closed valve configuration, wherein the fluid is prohibited from flowing through the fluid flushing system; and a backflow preventer disposed between the sampling port and the valve, the backflow preventer configured to allow the fluid to flow in a first direction through the fluid flushing system towards the valve and to prevent the fluid from flowing in an opposite second direction through the fluid flushing system towards the sampling port.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. application Ser. No. 16/930,962, filed Jul. 16, 2020, which is hereby specifically incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) This disclosure relates to flushing systems. More specifically, this disclosure relates to a remotely-operable flushing system for fluid distribution systems.

BACKGROUND

(2) Flushing systems can be used to periodically flush fluid from fluid systems, such as water systems. Flushing water systems can be done for a variety of reasons, including improving the quality of the water. Flushing systems are typically contained within a housing comprising a removable lid. To operate the flushing system, an operator must typically remove the lid from the housing and manually activate the flushing system. An operator must also typically be physically present to obtain various information related to fluid within the flushing system or the flushing system itself.

SUMMARY

(3) It is to be understood that this summary is not an extensive overview of the disclosure. This summary is exemplary and not restrictive, and it is intended neither to identify key or critical elements of the disclosure nor delineate the scope thereof. The sole purpose of this summary is to explain and exemplify certain concepts of the disclosure as an introduction to the following complete and extensive detailed description.

(4) Disclosed is a remote-operated flushing system comprising a fluid routing assembly comprising a valve, the valve configurable in an open configuration, wherein fluid is permitted to flow through the fluid routing assembly, and a closed configuration, wherein the fluid is prohibited from flowing through the fluid routing assembly; a control device configured to actuate the valve between the open configuration and closed configuration; a remote operation device wirelessly connected to the control device and configured to remotely operate the control device to control the actuation of the valve between the open configuration and closed configuration; and a sensor configured to detect a fluid property of the fluid within the fluid routing assembly, wherein the control device is configured to wirelessly send a signal representative of the fluid property detected by the sensor.

(5) Also disclosed is a remote-operated flushing system comprising a fluid routing assembly

comprising a valve, the valve configurable in an open configuration, wherein fluid is permitted to flow through the fluid routing assembly, and a closed configuration, wherein the fluid is prohibited from flowing through the fluid routing assembly; and a pressure monitoring system, the pressure monitoring system comprising: a pressure sensor mounted to the fluid routing assembly and configured to detect a pressure of the fluid within the fluid routing assembly; and a pressure monitoring unit configured to wirelessly send a pressure signal representative of the pressure detected by the pressure sensor, the pressure monitoring unit further configured to wirelessly receive a control signal from a remote operation device and to actuate the valve between the open configuration and closed configuration in response to the control signal.

(6) Also disclosed is a method of operating a flushing system, the method comprising providing a flushing system comprising a fluid routing assembly and a control device, the fluid routing assembly comprising a valve configurable in an open configuration, wherein fluid is permitted to flow through the fluid routing assembly, and a closed configuration, wherein the fluid is prohibited from flowing through the fluid routing assembly; remotely sending a control signal to a control device; actuating the valve between the open configuration and closed configuration with the control device in response to the control signal; detecting a pressure of the fluid with a pressure sensor; and sending a signal with the control device, the signal representative of the pressure detected by the pressure sensor.

(7) Additionally, disclosed is a fluid flushing system comprising a sampling port comprising a sampling conduit configured to dispense a sample of a fluid in the fluid routing assembly; a valve configurable in an open valve configuration, wherein the fluid is permitted to flow through the fluid flushing system, and a closed valve configuration, wherein the fluid is prohibited from flowing through the fluid flushing system; and a backflow preventer disposed between the sampling port and the valve, the backflow preventer configured to allow the fluid to flow in a first direction through the fluid flushing system towards the valve and to prevent the fluid from flowing in an opposite second direction through the fluid flushing system towards the sampling port.

(8) A method of operating a fluid flushing system is also disclosed, the method comprising providing the fluid flushing system comprising an inlet conduit, an outlet conduit, a backflow preventer disposed between the inlet conduit and the outlet conduit, and a flow valve disposed between the backflow preventer and the outlet conduit, wherein the backflow preventer is configured to allow a fluid to flow in a first direction through the fluid flushing system towards the flow valve and to prevent the fluid from flowing in an opposite second direction through the fluid flushing system towards the inlet conduit; actuating a shutoff valve of the backflow preventer to an open configuration to allow the fluid to flow through the backflow preventer from the inlet conduit to the flow valve; and actuating the shutoff valve to a closed configuration to prohibit fluid from flowing through the backflow preventer.

(9) Various implementations described in the present disclosure may include additional systems, methods, features, and advantages, which may not necessarily be expressly disclosed herein but will be apparent to one of ordinary skill in the art upon examination of the following detailed description and accompanying drawings. It is intended that all such systems, methods, features, and advantages be included within the present disclosure and protected by the accompanying claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features and components of the following figures are illustrated to emphasize the general principles of the present disclosure. Corresponding features and components throughout the figures may be designated by matching reference characters for the sake of consistency and clarity.

(2) FIG. 1A is a perspective view of a flushing system, in accordance with one aspect of the present

disclosure.

(3) FIG. 1B is a top view of the flushing system of FIG. 1A.

(4) FIG. 2 is a top perspective view of a fluid routing assembly of the flushing system of FIG. 1A, wherein the fluid routing assembly comprises backflow preventer and a valve.

(5) FIG. 3 is a rear perspective view of a pressure monitoring unit and a Bluetooth® controller of the flushing system of FIG. 1A.

(6) FIG. 4 is a top perspective view of a housing of the flushing system of FIG. 1A.

(7) FIG. 5 is a bottom perspective view of the housing of FIG. 4.

(8) FIG. 6 is a system diagram illustrating a method of operating the flushing system of FIG. 1A.

DETAILED DESCRIPTION

(9) The present disclosure can be understood more readily by reference to the following detailed description, examples, drawings, and claims, and the previous and following description. However, before the present devices, systems, and/or methods are disclosed and described, it is to be understood that this disclosure is not limited to the specific devices, systems, and/or methods disclosed unless otherwise specified, and, as such, can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting.

(10) The following description is provided as an enabling teaching of the present devices, systems, and/or methods in its best, currently known aspect. To this end, those skilled in the relevant art will recognize and appreciate that many changes can be made to the various aspects of the present devices, systems, and/or methods described herein, while still obtaining the beneficial results of the present disclosure. It will also be apparent that some of the desired benefits of the present disclosure can be obtained by selecting some of the features of the present disclosure without utilizing other features. Accordingly, those who work in the art will recognize that many modifications and adaptations to the present disclosure are possible and can even be desirable in certain circumstances and are a part of the present disclosure. Thus, the following description is provided as illustrative of the principles of the present disclosure and not in limitation thereof.

(11) As used throughout, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “an element” can include two or more such elements unless the context indicates otherwise.

(12) Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

(13) For purposes of the current disclosure, a material property or dimension measuring about X or substantially X on a particular measurement scale measures within a range between X plus an industry-standard upper tolerance for the specified measurement and X minus an industry-standard lower tolerance for the specified measurement. Because tolerances can vary between different materials, processes and between different models, the tolerance for a particular measurement of a particular component can fall within a range of tolerances.

(14) As used herein, the terms “optional” or “optionally” mean that the subsequently described event or circumstance can or cannot occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

(15) The word “or” as used herein means any one member of a particular list and also includes any combination of members of that list. Further, one should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain aspects include, while other aspects do not include, certain features, elements and/or steps. Thus, such conditional

language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular aspects or that one or more particular aspects necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular aspect.

(16) Disclosed are components that can be used to perform the disclosed methods and systems. These and other components are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these components are disclosed that while specific reference of each various individual and collective combinations and permutations of these may not be explicitly disclosed, each is specifically contemplated and described herein, for all methods and systems. This applies to all aspects of this application including, but not limited to, steps in disclosed methods. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific aspect or combination of aspects of the disclosed methods.

(17) Disclosed is a remote-operated flushing system and associated methods, systems, devices, and various apparatus. Example aspects of the remote-operated flushing system can comprise a valve configured to control fluid flow through the flushing system and a control device to allow remote operation of the valve. It would be understood by one of skill in the art that the flushing system is described in but a few exemplary embodiments among many. No particular terminology or description should be considered limiting on the disclosure or the scope of any claims issuing therefrom.

(18) FIG. 1A is a perspective view of a flushing system **100**, in accordance with one aspect of the present disclosure. The flushing system **100** can be configured to flush fluid, such as water, from a fluid system, such as, for example, a municipal water system, or any other fluid system where it may be desirable to periodically flush fluid out of the fluid system. For example, it may be desirable to flush stagnant or contaminated water out of the fluid system. As shown, example aspects of the flushing system **100** can comprise a housing **110** defining an interior cavity **115** within which various components of the flushing system **100** can be contained. In the present FIG. 1A, the housing **110** is illustrated as transparent for visibility of the components within the interior cavity **115**. According to example aspects, the housing **110** can comprise a sidewall enclosure **120** comprising a plurality of sidewalls **122** and defining an upper end **124** and a lower end **126**, relative to the orientation shown. In some aspects, an access opening **128** providing access to the interior cavity **115** can be formed at the upper end **124**. The housing **110** can further comprise a lid **425** (shown in FIG. 4) oriented at the upper end **124** of the sidewall enclosure **120** and a base **129** (shown in FIG. 1B) oriented at the lower end **126** of the sidewall enclosure **120**. As described in further detail below, in some aspects, the lid **425** can be configured to selectively uncover the access opening **128**, as shown, and cover the access opening **128**, as shown in FIG. 4. In example aspects, the housing **110** can be buried below or mostly below ground, and that the lid **425** can be oriented about flush with ground level. As such, the lid **425** can be removed from the sidewall enclosure **120** as needed, without the flushing system **100** extending above ground. Example aspects of the housing **110** can be formed from a composite material, a plastic material, such as polyvinyl chloride (PVC), a metal material, or any other suitable material or combination of materials known in the art.

(19) According to example aspects, the flushing system **100** can comprise a substantially U-shaped fluid routing assembly **130** configured to route fluid from the fluid system through the flushing system **100**. In other aspects, the fluid routing assembly **130** may not define a U-shape. Example aspects of the fluid routing assembly **130** can comprise a sampling port **135**, a backflow preventer **140**, and a valve **143**. In example aspects, the valve **143** can be an adjustable flow valve **145**, as shown, and can be configured to control the flow of fluid through the flushing system **100**. In other aspects, the valve **143** can be any other suitable type of valve known in the art. Additionally, in example aspects, some or all of the internal components of the adjustable flow valve **145** can

comprise a stainless steel material; however, in other aspects, the internal components of the adjustable flow valve **145** can comprise any other suitable material or combination of materials. An inlet pathway **150** can be provided for routing fluid into the fluid routing assembly **130** and an outlet pathway **160** can be provided for routing the fluid out of the fluid routing assembly **130**. In the present aspect, fluid from the fluid system can flow into the flushing system **100** through the inlet pathway **150**, past the sampling port **135**, through the backflow preventer **140**, through the adjustable flow valve **145**, and out of the flushing system **100** through the outlet pathway **160**. In some aspects, the fluid can further be configured to flow through a dechlorination unit **170**.

(20) As shown, the sampling port **135** can be oriented between the inlet pathway **150** and the backflow preventer **140**. According to example aspects, the sampling port **135** can comprise a sampling conduit **136** configured to dispense samples of the fluid in the fluid routing assembly **130** for testing the quality of the fluid. For example, the fluid can be tested for levels of lead, bacteria, nitrates, chlorine, pH levels, or the like. The fluid can be dispensed into a collection container and can be tested on site or taken to a testing facility. In some aspects, samples of the fluid can be obtained from the flushing system **100** even when the flushing system **100** is not actively flushing. In example aspects, the backflow preventer **140** can be oriented between the sampling port **135** and the adjustable flow valve **145**. The backflow preventer **140** can be configured to allow fluid to flow therethrough in a first direction towards the outlet pathway **160**, but can prevent the fluid from flowing in an opposite second direction back towards the inlet pathway **150**. As such, the fluid in the fluid system can be protected from contamination by the fluid in the flushing system **100** backflowing into the fluid system. In the present aspect, the backflow preventer **140** can be a double check valve **141**, though in other aspects, the backflow preventer **140** can define any other suitable configuration known in the art, such as an air gap. Example aspects of the backflow preventer **140** can comprise one or more relief valves **142**, as shown. The relief valves **142** may be spaced apart along the backflow preventer **140** to relieve air buildup within the backflow preventer **140** at various points.

(21) According to example aspects, the adjustable flow valve **145** can be oriented between the backflow preventer **140** and the outlet pathway **160**. The adjustable flow valve **145** can be configured to control the operation of the flushing system **100**. For example, the adjustable flow valve **145** can be selectively oriented in an open configuration and a closed configuration. In the open configuration, fluid can flow through the adjustable flow valve **145**, and the flushing system **100** can flush the fluid through the fluid routing assembly **130**. In the closed configuration, the fluid can be prevented from flowing through the adjustable flow valve **145**, and the flushing system **100** can thus be prevented flushing the fluid through the fluid routing assembly **130**. Example aspects of the adjustable flow valve **145** can be a solenoid-operated adjustable flow valve **145**, though in other aspects, the adjustable flow valve **145** can be a hydraulic adjustable flow valve, pneumatic adjustable flow valve, or any other suitable type of adjustable flow valve. In still other aspects, the valve **143** may not be an adjustable flow valve **145**, and can instead comprise any other suitable type of valve known in the art. In aspects of the adjustable flow valve **145** that are solenoid-operated, a solenoid **146** (shown in FIG. 1B) can be configured to selectively either admit or release pressure into a main chamber of the adjustable flow valve **145**, thus orienting the adjustable flow valve **145** in the closed and open configurations, respectively, in order to prevent or permit fluid flow through the adjustable flow valve **145**, respectively. In some aspects, the adjustable flow valve **145** can further be oriented in any suitable number of partially-open configurations to selectively regulate the fluid flow through the adjustable flow valve **145**. Example aspects of the adjustable flow valve **145** can further comprise a strainer therein configured to filter debris out the fluid flowing therethrough.

(22) Example aspects of the flushing system **100** can further comprise a control device **175** configured to allow an operator to remotely control the operation of the flushing system **100**, i.e., to remotely control the selective orientation of the adjustable flow valve **145** in the open, closed, and

partially-opened configurations, from a remote operation device **610** (shown in FIG. 6). In some aspects, such as the present aspect, the control device **175** can be completely wireless, while in other aspects, the control device **175** can be wired to the adjustable flow valve **145**. For example, as shown, the flushing system **100** can comprise one or both of a pressure monitoring system **180** and a Bluetooth® controller **190**, and one or both of the pressure monitoring system **180** and Bluetooth® controller **190** can serve as the control device **175**. In some aspects, the control device **175** can also be configured to control the speed at which the adjustable flow valve **145** opens and closes, which can aid in preventing water hammer. Bluetooth® is one example of short distance wireless communication protocols, and can be used to implement personal-area networks (PANs). In aspects wherein the Bluetooth® controller **190** is the control device **175** and the adjustable flow valve **145** is solenoid-operated, the Bluetooth® controller **190** can be connected to the solenoid by one or more wires **147** (shown in FIG. 1B). In other aspects, the Bluetooth® controller **190** configured to wirelessly actuate the solenoid **146**. Example aspects of the Bluetooth® controller **190** can also be wirelessly connected to the remote operation device **610**, which can allow an operator to remotely send signals to the Bluetooth® controller **190** from the remote operation device **610**. In other aspects, any other suitable wireless communication technique(s) may be implemented for remotely controlling the adjustable flow valve **145** with the control device **175**.

(23) The remote operation device **610** can be, for example, a mobile phone, tablet, computer, or the like. In example aspects, a program or app can be downloaded onto the remote operation device **610**, through which the operator can send signals to the Bluetooth® controller **190**. For example, the remote operation device **610** can be configured to send a control signal(s) **615** (shown in FIG. 6) to the Bluetooth® controller **190**, and the Bluetooth® controller **190** can actuate the adjustable flow valve **145** in response to the control signal(s) **615**. As such, an operator can remotely actuate the adjustable flow valve **145** with the remote operation device **610** in order to remotely operate the flushing system **100**. Thus, the adjustable flow valve **145** of the flushing system **100** does not need to be physically accessed by an operator in order to be operated. Furthermore, the operator may be able to operate the flushing system **100** at a distance from Bluetooth® controller **190**. For example, an operator may be able to operate the flushing system **100** from across the street from the flushing system **100**. This can be beneficial in various instances, such as, for example, when the weather is poor and the operator wishes to stay inside their vehicle, or if the flushing system **100** is located in an area that is difficult to access. As shown in the present aspect, the Bluetooth® controller **190** can be mounted to the sidewall enclosure **120** of the housing **110**, though in other aspects, the Bluetooth® controller **190** can be mounted at any suitable location within the interior cavity **115**, including mounted to the fluid routing assembly **130**, the lid **425**, or the base **129**. Some aspects of the Bluetooth® controller **190** may be configured to control various other features of the flushing system **100** and/or may be configured to communicate information, such as water quality information, to one or more external electronic device(s) **620** (shown in FIG. 6). In a particular example aspect, the external electronic device(s) **620** can be or include a computer at a remote operations center. Furthermore, in some aspects, the external electronic device(s) **620** can be or can include the remote operation device **610**.

(24) According to some example aspects, the flushing system **100** can also or alternatively comprise the pressure monitoring system **180**. In some aspects, the pressure monitoring system **180** can be similar to the monitoring device disclosed in U.S. patent application Ser. No. 15/171,722, filed Jun. 2, 2016, which is hereby specifically incorporated by reference herein in its entirety. Example aspects of the pressure monitoring system **180** can comprise a pressure sensor **182** (shown in FIG. 1B) and a pressure monitoring unit **185**. The pressure sensor **182** can be, for example, a piezo-resistive strain gauge, a capacitive gauge, an electromagnetic gauge, a piezoelectric device, or any other suitable device known in the art for detecting pressure. The pressure sensor **182** can be mounted within the fluid routing assembly **130** such that the pressure sensor **182**, or a portion thereof, is in contact with the fluid therein. The pressure sensor **182** can be wired to the pressure

monitoring unit **185** by one or more wires **183** (shown in FIG. 1B) and can transmit pressure data obtained by the pressure sensor **182** through the wires **183** to the pressure monitoring unit **185**. In other aspects, the pressure sensor **182** may be configured to wirelessly transmit the pressure data to the pressure monitoring unit **185**. In various aspects, the pressure sensor **182** can be configured to continually communicate pressure data to the pressure monitoring unit **185**, while in other aspects, the pressure sensor **182** can communicate pressure data periodically or only when an anomaly is detected. The pressure monitoring unit **185** can be configured to evaluate the pressure data to determine whether a concern is present. For example, the pressure monitoring unit **185** may comprise a printed circuit board or other processing unit configured to process and evaluate the pressure data. Example aspects of the pressure monitoring unit **185** can also be configured to send a pressure signal **685** (shown in FIG. 6) representative of the detected pressure to a designated electronic device(s) **620**, which may include the remote operation device **610**, as described in further detail below. In instances wherein the pressure data presents a concern, the pressure monitoring unit **185** may be configured to send an alert signal to the designated electronic device(s) **620**. Furthermore, in some aspects, the pressure monitoring unit **185** can be the control device **175** and can be configured to control the operation of the solenoid **146**, and thus, the Bluetooth® controller **190** may not be required. As shown, the pressure monitoring unit **185** can be connected to the solenoid **146** by one or more wires **148**, or may be wirelessly connected to the solenoid **146**. In example aspects, the solenoid **146** can be connected to only one of the Bluetooth® controller **190** and the pressure monitoring unit **185**, though in other aspects, the solenoid **146** may be connected to both. Moreover, in some aspects, the pressure monitoring unit **185** may be configured to control various other features of the flushing system **100** and/or may be configured to communicate information, such as water quality information, to the designated external electronic device(s) **620**. As shown in the present aspect, the pressure monitoring unit **185** can be mounted to the sidewall enclosure **120** of the housing **110**, though in other aspects, the pressure monitoring unit **185** can be mounted at any suitable location within the interior cavity **115**, including mounted to the fluid routing assembly **130**, the lid **425**, or the base **129**.

(25) According to example aspects, the flushing system **100** can further comprise the dechlorination unit **170** received within the housing **110**. In some aspects, the dechlorination unit **170** can be mounted to the housing **110**, for example, to the sidewall enclosure **120**, and in other aspects, the dechlorination unit **170** can be mounted to the fluid routing assembly **130**. The dechlorination unit **170** can be configured to dechlorinate fluid as it flows therethrough. For example, in some aspects, the dechlorination unit **170** can comprise dechlorination tablets therein which can dechlorinate the fluid as the fluid passes over the dechlorination tablets. The dechlorination tablets can comprise sodium sulfite, ascorbic acid, or any other suitable substance for dechlorinating fluid. According to example aspects, some or all of the fluid being flushed through the flushing system **100** can be routed through the dechlorination unit **170** for dechlorination. As shown, the fluid can be transferred from the fluid routing assembly **130** to the dechlorination unit **170** through a dechlorination inlet conduit **172**, and, once dechlorinated, the fluid can be transferred from the dechlorination unit **170** back to the fluid routing assembly **130** through a dechlorination outlet conduit **174**. In the present aspect, the dechlorination inlet conduit **172** can extend from the valve **143** to the dechlorination unit **170**, and the dechlorination outlet conduit **174** can extend from the dechlorination unit **170** to the outlet pathway **160**, downstream of the valve **143**. In some aspects, the dechlorination unit **170** may comprise a dechlorination valve that can be selectively adjusted to control the rate of dechlorination. According to example aspects, various local, state, or national standards may exist for the dechlorination of fluid flushed from a fluid system, and the dechlorination unit **170** can be designed to meet or exceed these standards.

(26) FIG. 1B illustrates a top view of the flushing system **100**, wherein the lid **425** (shown in FIG. 4) is removed for visibility into the interior cavity **115**. As shown, the pressure sensor **182** can be mounted to the valve **143** at an, and the pressure monitoring unit **185** can be connected to the

pressure sensor **182** by the wire **183**. The pressure monitoring unit **185** can also be connected to the solenoid **146** by the wire **148**. Some aspects of the flushing system **100** can also or alternatively include the Bluetooth® controller **190**, which can be connected to the solenoid **146** by the wires **147**. In other aspects, the pressure monitoring unit **185** may be wirelessly connected to either or both of the pressure sensor **182** and the solenoid **146** and/or the Bluetooth® controller **190** may be wirelessly connected to the solenoid **146**.

(27) FIG. 2 illustrates a top perspective view of the fluid routing assembly **130**. As shown, the inlet pathway **150** can comprise an inlet conduit **252** that can be connected to the fluid system and can provide a path for the fluid in the fluid system to enter the fluid routing assembly **130**. The inlet conduit **252** can be configured to extend into the interior cavity **115** through an inlet opening **532** (shown in FIG. 5) formed in the base **129** (shown in FIG. 1B) of the housing **110** (shown in FIG. 1A). In some aspects, an inlet mounting bracket **253** can be mounted to the inlet conduit **252**, and the inlet mounting bracket **253** can be attached to the base **129** to secure the inlet conduit **252** to the housing **110**. As shown, one or more fasteners, such as nut and bolt fasteners **255**, may be provided for securing the inlet mounting bracket **253** to the base **129**. In example aspects, the inlet pathway **150** can be configured to extend substantially upward, relative to the orientation shown, towards the upper end **124** (shown in FIG. 1A) of the sidewall enclosure **120** (shown in FIG. 1A). As shown, in the present aspect, an inlet connector **254** can be provided for connecting the inlet conduit **252** to an inlet pipe **256**. In some aspects, one or both of the inlet conduit **252** and inlet pipe **256** can be threadably coupled to the inlet connector **254**. An inlet elbow fitting **258** can be coupled to the inlet pipe **256** and can define a bend angle of about 90°. As such, the fluid can flow from the fluid system into the inlet conduit **252**, and can then flow through the inlet connector **254**, inlet pipe **256**, and inlet elbow fitting **258**. Example aspects of the inlet pathway **150** can comprise more or fewer components as needed to route the fluid to the backflow preventer **140**.

(28) Example aspects of the outlet pathway **160** can comprise an outlet conduit **262** that can provide a path for the fluid to exit the fluid routing assembly **130**. In example aspects, the outlet conduit **262** can be connected to a fluid discharge location, such as sewage system, storm system, swale, retention system, or the like. In some aspects, as described above, the fluid flushed through the flushing system **100** can be routed through the dechlorination unit **170** (shown in FIG. 1A) prior to being discharged at the fluid discharge location. The outlet conduit **262** can be configured to extend into the interior cavity **115** through an outlet opening **534** (shown in FIG. 5) formed in the base **129** (shown in FIG. 1B) of the housing **110** (shown in FIG. 1A). In some aspects, an outlet mounting bracket **263** can be mounted to the outlet conduit **262**, and the outlet mounting bracket **263** can be attached to the base **129** to secure the outlet conduit **262** to the housing **110**. As shown, one or more fasteners, such as nut and bolt fasteners **265**, may be provided for securing the outlet mounting bracket **263** to the base **129**. According to some example aspects, the outlet pathway **160** can be configured to extend substantially upward, relative to the orientation shown, towards the upper end **124** of the sidewall enclosure **120**. In the present aspect, the outlet pathway **160** can further comprise an outlet connector **264** for connecting the outlet conduit **262** to an outlet pipe **266**. In some aspects, the outlet conduit **262** and outlet pipe **266** can be threadably coupled to the outlet connector **264**. As such, as shown, the fluid in the flushing system **100** can exit the flushing system **100** by flowing into the outlet pipe **266**, and then through the outlet connector **264** and the outlet conduit **262**. Example aspects of the outlet pathway **160** can comprise more or fewer components as needed to route the fluid out of the flushing system **100**. In other aspects, either or both of the inlet pathway **150** and outlet pathway **160** comprise any suitable configuration for routing the fluid into and out of the flushing system **100**.

(29) According to example aspects, a backflow preventer inlet **210** can be oriented between and coupled to the inlet elbow fitting **258** and to an inlet end **242** of the backflow preventer **140**, such that fluid can flow from the inlet elbow fitting **258**, through the backflow preventer inlet **210**, and into the backflow preventer **140**. In some aspects, the backflow preventer inlet **210** can be

threadably coupled to the inlet elbow fitting **258**. As shown, the sampling port **135** extend from and can be in fluid communication with the backflow preventer inlet **210**. As such, the sampling port **135** can be oriented proximate to the upper end **124** of the sidewall enclosure **120**, such that the sampling port **135** can be easily accessed when the lid **425** is removed from the housing **110** to allow access to the interior cavity **115**. In other aspects, however, the sampling port **135** can be oriented at any other suitable location in the flushing system **100**. Additionally, as shown, the backflow preventer inlet **210** can comprise an inlet shutoff valve **212**, such as a ball valve, which, in the present aspect, can be manually operated to selectively shut off fluid flow into the backflow preventer **140**.

(30) Example aspects of the fluid routing assembly **130** can further comprise a backflow preventer outlet **220** oriented between and coupled to an outlet end **244** of the backflow preventer **140** and an inlet end **246** of the adjustable flow valve **145**. Thus, fluid can be configured to flow from the backflow preventer **140**, through the backflow preventer outlet **220**, and into the adjustable flow valve **145**. Furthermore, as shown, the outlet pipe **266** of the outlet pathway **160** can be connected to an outlet end **248** of the adjustable flow valve **145**, such that fluid can flow out of the adjustable flow valve **145** at the outlet end **248** and into the outlet pathway **160**. In some aspects, the backflow preventer outlet **220** can be threadably coupled to a threaded valve connector **224**, and the threaded valve connector **224** can be threadably coupled to the inlet end **246** of the adjustable flow valve **145**. Similar to the backflow preventer inlet **210**, the backflow preventer outlet **220** can comprise an outlet shutoff valve **222**, such as a ball valve, which can be manually operated to selectively shut off fluid flow out of the backflow preventer **140**. In other aspects, the inlet and outlet shutoff valves **212**, **222** may not be manually-operated, and may instead be automatically operated. Each of the backflow preventer **140** and adjustable flow valve **145** can be oriented proximate to the upper end **124** of the sidewall enclosure **120**, to allow for easy access thereto through the access opening **128** for operation of the inlet and outlet shutoff valves **212**, **222** or for the repair or replacement of parts. Furthermore, according to some example aspects, the backflow preventer inlet **210** can be coupled to the backflow preventer outlet **220** by one or more connecting brackets **230**. For example, as shown, the backflow preventer inlet **210** can be coupled to the backflow preventer outlet **220** by a first connecting bracket **230a** and a second connecting bracket **230b**. In the present view, the second connecting bracket **230b** is largely hidden from view by the backflow preventer **140**, but can be substantially the same as the first connecting bracket **230a**.

(31) FIG. 3 illustrates the Bluetooth® controller **190** and pressure monitoring unit **185** mounted to the sidewall enclosure **120** of the housing **110**. The sidewall enclosure **120** is again illustrated as transparent for visibility into the interior cavity **115**. As described above, other aspects of the flushing system **100** may comprise only one of the Bluetooth® controller **190** and the pressure monitoring unit **185**. According to example aspects, the pressure monitoring system **180** can comprise an antenna configured to send pressure signals **685** (shown in FIG. 6) representative of the pressure data received from the pressure sensor **182** (shown in FIG. 1B) to a designated external electronic device(s) **620** (shown in FIG. 6), which as described, may be or may include the remote operation device **610**. Optionally, the housing **110** can be formed from a non-ferrous material, so that the material does not interfere with the ability of the antenna to send the pressure signals **685** and other signals externally. In the present aspect, the antenna can be configured to send signals, including the pressure signals **685**, over a cellular network. However, in other aspects, signals can be sent from the pressure monitoring unit **185** over wifi, ethernet, Bluetooth®, or any other suitable wireless technology. The pressure monitoring unit **185** may be configured to report the pressure data externally continually, at user-defined intervals, or may be configured to report the pressure data solely when an anomaly occurs, such as a spike in pressure.

(32) In some aspects, the antenna can also allow an operator to remotely control the operation of the flushing system **100**. For example, in aspects comprising the solenoid-operated adjustable flow valve **145**, the antenna can allow the operator to wirelessly control the solenoid **146** (shown in FIG.

1B) to selectively orient the adjustable flow valve **145** in the open, closed, and partially-open configurations. As described above, the antenna can be wirelessly connected to an app or program on the remote operation device **610** through which the operator can control the flushing system **100**. In such aspects, it may not be necessary to include the Bluetooth® controller **190** in the flushing system **100**. Furthermore, in some aspects, the adjustable flow valve **145** can also be programmed to automatically open and close periodically for routine flushing of the fluid system. In example aspects, when the adjustable flow valve **145** moves to the open or partially open configuration, allowing fluid to flow through the flushing system **100**, the pressure of the fluid can drop and a pressure signal **685** can be sent by the antenna relaying the pressure drop information to the designated external electronic device(s) **620**. In some aspects, the external electronic device(s) **620** can be or can include the remote operation device **610**. Similarly, the pressure of the fluid can increase when the adjustable flow valve **145** moves to the closed configuration, and a pressure signal **685** can be sent relaying the pressure increase information to the designated external electronic device(s) **620**.

(33) According to example aspects, the antenna may allow for remote control of various other features of the flushing system **100**. Furthermore, in addition to pressure information, the antenna may be configured to communicate information related to other aspects of the flushing system **100** or the fluid therein to one or more external electronic device(s) **620**. For example, the flushing system **100** may comprise a temperature sensor **184** configured to detect a temperature of the fluid and the antenna can be configured to send a temperature signal representative of the detected temperature to the designated electronic device(s) **620**. In the present aspect, the temperature sensor **184** can be housed with the pressure sensor **182** and can be wired to the pressure monitoring unit **185** by the wire **183**. Example aspects of the flushing system **100** may further comprise various other sensors, detectors, and/or measurement tools for sensing, detecting, and/or measuring other properties of the fluid, such as, for example, fluid quality, flow rate, pH level, chlorine level, disinfectant level, turbidity, and the like. The antenna can communicate information related to detected fluid property or properties via a fluid property signal. The antenna can also be configured to communicate information such as the concentration of the dechlorination substances (e.g., sodium sulfite, ascorbic acid) within the dechlorination unit **170**, a status of the strainer in the adjustable flow valve **145**, etc. In aspects wherein the Bluetooth® controller **190** is also provided, the Bluetooth® controller **190** may allow for remote control the same or different features of the flushing system **100** and/or communication of the same or different information.

(34) According to example aspects, the pressure monitoring unit **185** can be mounted to the sidewall enclosure **120** by a pressure monitor bracket **310** and the Bluetooth® controller **190** can be mounted to the sidewall enclosure **120** by one or more mounting tabs **320**. In the present aspect, the Bluetooth® controller **190** can be mounted to a first one of the sidewalls **122a** of the sidewall enclosure **120** and the pressure monitoring unit **185** can be mounted to a second one of the sidewalls **122b**. In other aspects, the Bluetooth® controller **190** and sidewall enclosure **120** can be mounted at any other suitable location within the interior cavity **115**, including any location on the sidewall enclosure **120**, lid **425** (shown in FIG. 4), base **129** (shown in FIG. 1B), and fluid routing assembly **130** (shown in FIG. 1A). As shown, example aspects of the pressure monitor bracket **310** comprise a bracket ring **312** and a bracket mounting flange **314**. The bracket ring **312** can wrap around a body **386** of the pressure monitoring unit **185** and a head **388** of the pressure monitoring unit **185** can rest on the bracket ring **312**. The bracket mounting flange **314** can abut the second sidewall **122b** of the sidewall enclosure **120** and can define one or more fastener holes **316** formed therethrough, as shown. A fastener such as, for example, a screw, can extend through each of the fastener holes **316** and can engage the second sidewall **122b** to mount the pressure monitor bracket **310** to the sidewall enclosure **120**. Example aspects of the Bluetooth® controller **190** can comprise two of the mounting tabs **320** formed monolithically with the Bluetooth® controller **190**, and the mounting tabs **320** can abut the first sidewall **122a** of the sidewall enclosure **120**. In other aspects,

the mounting tabs **320** can be formed separately from the Bluetooth® controller **190** and attached thereto. Each of the mounting tabs **320** can define a fastener hole **326** formed therethrough, and a fastener, such as a screw, for example, can extend through each of the fastener holes **326** of the mounting tabs **320** and can engage the first sidewall **122a** to mount the Bluetooth® controller **190** to the sidewall enclosure **120**. In other aspects, the Bluetooth® controller **190** and pressure monitoring unit **185** can be secured to the housing **110** or elsewhere in the interior cavity **115** by any other suitable attachments mechanisms known in the art.

(35) FIG. 4 illustrates a top perspective view of the housing **110** with the lid **425** covering access opening **128** (shown in FIG. 1A) to prevent access to the interior cavity **115** (shown in FIG. 1A). In example aspects, the lid **425** may be configured to rest on a rim or stops of the sidewall enclosure **120** to support the lid **425** proximate to the upper end **124** of the sidewall enclosure **120**. In other aspects, a friction fit may be defined between the lid **425** and sidewall enclosure **120** to secure the lid **425** in position at the upper end **124**, and in still other aspects, the housing **110** can define any other suitable configuration for retaining the lid **425** at the upper end **124** of the sidewall enclosure **120**. As described above, in example aspect, the lid **425** may be removable from the sidewall enclosure **120** to uncover the access opening **128** and allow access to the interior cavity **115**. For example, the lid **425** may be removed for the manual operation of the inlet and outlet shutoff valves **212**, **222** (shown in FIG. 2), for obtaining fluid samples from the sampling port **135** (shown in FIG. 1A), for replacing the strainer in the adjustable flow valve **145** (shown in FIG. 1A), for adding additional dechlorination tablets to the dechlorination unit **170** shown in FIG. 1A), for repairing or replacing any of the components housed within the interior cavity **115**, or for any other suitable reason. Example aspects of the lid **425** may comprise a handle or handles to facilitate lifting the lid **425** away from the sidewall enclosure **120**. Moreover, in some aspects, a tool may be required to remove the lid **425** in order to prohibit manual removal of the lid **425** and prevent unintentional removal of the lid **425** and/or tampering with the flushing system **100**. In other aspects, the lid **425** may not be removable from housing **110**.

(36) FIG. 5 illustrates a bottom perspective view of the housing **110**, illustrating the base **129** oriented at the lower end **126** of the sidewall enclosure **120**. In some aspects, the base **129** can be monolithically formed with sidewall enclosure **120**, and in other aspects, the base **129** can be separately formed from the sidewall enclosure **120** and attached thereto. As such, some example aspects of the base **129** can be removable from the sidewall enclosure **120**, while in other aspects, the base **129** is not removable. As shown, example aspects of the base **129** can define the inlet opening **532** formed therethrough through which the inlet conduit **252** (shown in FIG. 2) can extend and the outlet opening **534** formed therethrough through which the outlet conduit **262** (shown in FIG. 2) can extend. Other aspects of the base **129** may comprise additional openings to allow additional components of the flushing system **100**, or other systems, to extend into and/or out of the interior cavity **115** (shown in FIG. 1A).

(37) FIG. 6 is a system diagram illustrating a method of operating the flushing system **100**, according to an example aspect of the disclosure. Example aspects of the method can include providing the flushing system **100** comprising the fluid routing assembly **130** (shown in FIG. 1A) and the control device **175**, wherein the fluid routing assembly **130** can comprise the valve **143**. The valve can be configurable in the open configuration, wherein fluid is permitted to flow through the fluid routing assembly **130**, and a closed configuration, wherein the fluid is prohibited from flowing through the fluid routing assembly **130**. The method can further comprising remotely sending a control signal **615** to the control device **175**. For example, the control signal **615** can be sent by the remote operation device **610**. Example aspects of the method can also comprise actuating the valve **143** between the open configuration and closed configuration with the control device **175** in response to the control signal **615**, wherein, in some aspects, the control device **175** can wirelessly actuate the valve **143** by sending an actuation signal **645** to the valve **143**. In some aspects, the method can further comprise detecting a pressure of the fluid with the pressure sensor

182, and sending the pressure signal **685** with the control device **175**, the pressure signal **685** representative of the pressure detected by the pressure sensor **182**. According to example aspects, the pressure signal **685** can be sent to an external electronic device **620** or devices, and in some aspects, the external electronic device(s) **620** can be or can comprise the remote operation device **610**. In example aspects, the pressure sensor **182** can send pressure data wirelessly to the control device **175**, while in other aspects, the pressure sensor **182** can be wired to the control device **175** and can transmit data to the control device **175** through a pressure sensor wire **640**, as shown.

(38) One should note that conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more particular embodiments or that one or more particular embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment.

(39) It should be emphasized that the above-described embodiments are merely possible examples of implementations, merely set forth for a clear understanding of the principles of the present disclosure. Any process descriptions or blocks in flow diagrams should be understood as representing modules, segments, or portions of code which include one or more executable instructions for implementing specific logical functions or steps in the process, and alternate implementations are included in which functions may not be included or executed at all, may be executed out of order from that shown or discussed, including substantially concurrently or in reverse order, depending on the functionality involved, as would be understood by those reasonably skilled in the art of the present disclosure. Many variations and modifications may be made to the above-described embodiment(s) without departing substantially from the spirit and principles of the present disclosure. Further, the scope of the present disclosure is intended to cover any and all combinations and sub-combinations of all elements, features, and aspects discussed above. All such modifications and variations are intended to be included herein within the scope of the present disclosure, and all possible claims to individual aspects or combinations of elements or steps are intended to be supported by the present disclosure.

Claims

1. A fluid flushing system comprising: a sampling port comprising a sampling conduit configured to dispense a sample of a fluid in the fluid flushing assembly; a valve configurable in an open valve configuration, wherein the fluid is permitted to flow through the fluid flushing system, and a closed valve configuration, wherein the fluid is prohibited from flowing through the fluid flushing system; a backflow preventer disposed between the sampling port and the valve, the backflow preventer configured to allow the fluid to flow in a first direction through the fluid flushing system towards the valve and to prevent the fluid from flowing in an opposite second direction through the fluid flushing system towards the sampling port; a housing comprising a base and defining an interior cavity, wherein the sampling port, the valve, and the backflow preventer are oriented within the housing; and an inlet pathway configured to route the fluid from a fluid system into the fluid flushing system and an outlet pathway configured to route the fluid out of the fluid flushing system; wherein: the sampling port is disposed between the inlet pathway and the backflow preventer; the valve is disposed between the backflow preventer and the outlet pathway; the inlet pathway comprises an inlet conduit extending into the interior cavity through an inlet opening in the base; and the outlet pathway comprises an outlet conduit extending into the interior cavity through an outlet opening in the base.

2. The fluid flushing system of claim 1, further comprising a pressure monitoring system, the

- pressure monitoring system comprising a pressure sensor mounted to the fluid flushing system and configured to detect a pressure of the fluid in the fluid flushing system.
3. The fluid flushing system of claim 2, wherein the pressure monitoring system further comprises a pressure monitoring unit wired to the pressure sensor, the pressure monitoring unit comprising a processing unit and an antenna, the processing unit configured to process pressure data from the pressure sensor and to relay the pressure data to the antenna, the antenna configured to send a pressure signal representative of the pressure detected by the pressure sensor.
4. The fluid flushing system of claim 3, wherein the antenna is further configured to wirelessly receive a control signal from a remote operation device and to actuate the valve between the open valve configuration and the closed valve configuration in response to the control signal.
5. The fluid flushing system of claim 4, further comprising a remote operation device wirelessly connected to the valve and configured to actuate the valve between the open valve configuration and the closed valve configuration.
6. The fluid flushing system of claim 3, further comprising a temperature sensor configured to detect a temperature of the fluid in the fluid flushing system, and wherein the antenna is configured to send a temperature signal representative of the temperature detected by the temperature sensor.
7. The fluid flushing system of claim 1, further comprising an inlet mounting bracket mounted to the inlet conduit and an outlet mounting bracket mounted to the outlet conduit, the inlet mounting bracket attached to the base to secure the inlet conduit to the housing, the outlet mounting bracket attached to the base to secure the outlet conduit to the housing.
8. The fluid flushing system of claim 1, wherein: the housing further comprises a sidewall enclosure and a lid: the sidewall enclosure defines an access opening allowing access to the interior cavity; and the lid is configurable in a closed lid configuration, wherein the lid covers the access opening, and an open lid configuration, wherein the lid is removed from the housing and the access opening is uncovered to permit access to the interior cavity.
9. The fluid flushing system of claim 8, wherein the access opening is formed at an upper end of the sidewall enclosure, and wherein the sampling port is oriented proximate to the upper end of the sidewall enclosure to facilitate accessing the sampling port through the access opening.
10. The fluid flushing system of claim 1, wherein: the backflow preventer defines a backflow preventer inlet and a backflow preventer outlet; the backflow preventer comprises an inlet shutoff valve at the backflow preventer inlet operable to prevent the fluid from flowing into the backflow preventer; and the backflow preventer comprises an outlet shutoff valve at the backflow preventer outlet operable to prevent the fluid from flowing out of the backflow preventer.
11. The fluid flushing system of claim 10, wherein the backflow preventer inlet is coupled to the backflow preventer outlet by a backflow connecting bracket.
12. The fluid flushing system of claim 10, wherein the backflow preventer comprises a plurality of relief valves spaced apart along the backflow preventer and configured to relieve air buildup within the backflow preventer.
13. A method of operating a fluid flushing system comprising: providing the fluid flushing system comprising a housing comprising a base and defining an interior cavity, an inlet conduit extending into the interior cavity through an inlet opening defined in the base, an outlet conduit extending into the interior cavity through an outlet opening defined in the base, a backflow preventer disposed between the inlet conduit and the outlet conduit, and a flow valve disposed between the backflow preventer and the outlet conduit, wherein the backflow preventer is configured to allow a fluid to flow in a first direction through the fluid flushing system towards the flow valve and to prevent the fluid from flowing in an opposite second direction through the fluid flushing system towards the inlet conduit; actuating a shutoff valve of the backflow preventer to an open configuration to allow the fluid to flow through the backflow preventer from the inlet conduit to the flow valve; and actuating the shutoff valve to a closed configuration to prohibit fluid from flowing through the backflow preventer.

14. The method of claim 13, wherein: actuating the shutoff valve of the backflow preventer to the open configuration comprises manually actuating the shutoff valve to the open configuration; and actuating the shutoff valve to the closed configuration comprises manually actuating the shutoff valve to the closed configuration.

15. The method of claim 13, further comprising actuating the flow valve from a closed flow configuration to an open flow configuration when the shutoff valve of the backflow preventer is in the open configuration.

16. The method of claim 13, further comprising releasing air buildup within the backflow preventer through a relief valve of the backflow preventer.

17. The method of claim 13, wherein: the backflow preventer defines a backflow preventer inlet and a backflow preventer outlet; the shutoff valve is an inlet shutoff valve at the backflow preventer inlet that is operable to prevent the fluid from flowing into the backflow preventer from the inlet conduit; and the backflow preventer further comprises an outlet shutoff valve at the backflow preventer outlet operable to prevent the fluid from flowing out of the backflow preventer to the flow valve.
